

The Unregularized Nagumo Inverse Problem

Weighted Least Squares, Fisher Information, and Confidence Intervals

Mathematical note

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Abstract

This note gives a mathematical description of a two-parameter inverse problem for the anisotropic Nagumo model. The diffusion coefficients are estimated by a box-constrained, noise-weighted nonlinear least-squares functional without a regularization term. Local parameter uncertainty is quantified through central finite-difference sensitivities, the Fisher information matrix, and asymptotic 95% confidence intervals. A numerical example from the current synthetic-data experiment is included.

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1 Parameter-to-observation map

Let

$$\theta = (a_1, a_2)^\top \in \Theta, \quad \Theta = [\ell_1, u_1] \times [\ell_2, u_2] \subset \mathbb{R}_{>0}^2, \quad (1)$$

where a_1 and a_2 are the two diagonal entries of the anisotropic diffusion tensor. For a fixed initial condition, sensor configuration, and set of observation times, the solution of the forward Nagumo model defines a nonlinear parameter-to-observation map

$$\mathcal{F} : \Theta \longrightarrow \mathbb{R}^m, \quad m = N_t N_s. \quad (2)$$

There are $N_t = 1000$ observation times and $N_s = 5$ sensors in the present experiment, so $m = 5000$. The vector $\mathcal{F}(\theta)$ is obtained by stacking all sensor values at the first time, followed by all sensor values at the second time, and so forth.

The statistical observation model is

$$y^{\text{obs}} = \mathcal{F}(\theta_{\text{true}}) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_m), \quad (3)$$

where the errors are independent, have zero mean, and share the known standard deviation $\sigma = 0.005$. Consequently,

$$y^{\text{obs}} \mid \theta \sim \mathcal{N}(\mathcal{F}(\theta), \sigma^2 I_m). \quad (4)$$

2 Noise-weighted inverse functional

The likelihood of observing y^{obs} at a candidate parameter θ is

$$p(y^{\text{obs}} \mid \theta) = (2\pi\sigma^2)^{-m/2} \exp\left[-\frac{1}{2\sigma^2} \|\mathcal{F}(\theta) - y^{\text{obs}}\|_2^2\right]. \quad (5)$$

Its negative logarithm is

$$-\log p(y^{\text{obs}} \mid \theta) = \frac{m}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \|\mathcal{F}(\theta) - y^{\text{obs}}\|_2^2. \quad (6)$$

Because σ is fixed, the first term is constant with respect to θ . Maximum likelihood estimation is therefore equivalent to

$$\boxed{\hat{\theta} = \underset{\theta \in \Theta}{\operatorname{argmin}} \Phi(\theta), \quad \Phi(\theta) = \frac{1}{2\sigma^2} \|\mathcal{F}(\theta) - y^{\text{obs}}\|_2^2.} \quad (7)$$

Equivalently, define the whitened residual

$$r(\theta) = \frac{\mathcal{F}(\theta) - y^{\text{obs}}}{\sigma}; \quad \Phi(\theta) = \frac{1}{2} \|r(\theta)\|_2^2. \quad (8)$$

There is no Tikhonov penalty, prior mean, or other regularization contribution in (7). Thus all curvature used to identify a_1 and a_2 comes from the observations through $\mathcal{F}$. The box Θ restricts the admissible parameter values but does not add a term to the functional.

3 Sensitivity matrix

The local response of the observations to the parameters is described by

$$S(\theta) = \frac{\partial \mathcal{F}(\theta)}{\partial \theta} = \begin{bmatrix} s_1(\theta) & s_2(\theta) \end{bmatrix} \in \mathbb{R}^{m \times 2}, \quad s_j(\theta) = \frac{\partial \mathcal{F}(\theta)}{\partial a_j}. \quad (9)$$

When analytic tangent or adjoint derivatives are unavailable, each column can be approximated by a central finite difference,

$$s_j(\hat{\theta}) \approx \frac{\mathcal{F}(\hat{\theta} + h_j e_j) - \mathcal{F}(\hat{\theta} - h_j e_j)}{2h_j}, \quad (10)$$

where e_j is the j th Euclidean coordinate vector. If $\eta > 0$ is a prescribed relative perturbation, a scale-aware requested step is

$$h_j^{(0)} = \eta \max(1, |\hat{\theta}_j|). \quad (11)$$

Central differences require both perturbed parameters to remain admissible. Let

$$d_j = \min(\hat{\theta}_j - \ell_j, u_j - \hat{\theta}_j). \quad (12)$$

A symmetric bound-aware choice is then

$$h_j = \min\left(h_j^{(0)}, \frac{d_j}{2}\right). \quad (13)$$

The factor $1/2$ keeps both perturbations strictly inside the box whenever $d_j > 0$. If d_j is at roundoff scale, a meaningful central derivative cannot be formed. More generally, h_j must balance the $O(h_j^2)$ truncation error of (10) against cancellation error from taking h_j too small.

4 Fisher information

For model (3), the score is

$$\nabla_{\theta} \log p(y^{\text{obs}} | \theta) = \frac{1}{\sigma^2} S(\theta)^{\top} (y^{\text{obs}} - \mathcal{F}(\theta)). \quad (14)$$

The Fisher information is the expected outer product of the score, or equivalently the negative expected Hessian of the log-likelihood. Using $\text{Cov}(y^{\text{obs}} - \mathcal{F}(\theta)) = \sigma^2 I_m$ gives

$$\boxed{\mathcal{I}(\theta) = \frac{1}{\sigma^2} S(\theta)^{\top} S(\theta)}. \quad (15)$$

Writing $S = [s_1 \ s_2]$ makes the two-parameter structure explicit:

$$\mathcal{I}(\theta) = \frac{1}{\sigma^2} \begin{pmatrix} s_1^{\top} s_1 & s_1^{\top} s_2 \\ s_2^{\top} s_1 & s_2^{\top} s_2 \end{pmatrix}. \quad (16)$$

The diagonal entries measure the information about each parameter separately, whereas the off-diagonal entry measures coupling between their observation sensitivities.

To relate this matrix to the functional, define $q(\theta) = \mathcal{F}(\theta) - y^{\text{obs}}$. Direct differentiation yields

$$\nabla \Phi(\theta) = \frac{1}{\sigma^2} S(\theta)^{\top} q(\theta), \quad (17)$$

$$\nabla^2 \Phi(\theta) = \frac{1}{\sigma^2} S(\theta)^{\top} S(\theta) + \frac{1}{\sigma^2} \sum_{i=1}^m q_i(\theta) \nabla^2 \mathcal{F}_i(\theta). \quad (18)$$

The first term is the Gauss–Newton Hessian and equals the Fisher information. At the true parameter the second term has zero expectation because the observation errors have zero mean. Hence $\mathcal{I}$ represents the expected local curvature supplied by the data, while the observed Hessian may also contain residual-weighted second derivatives.

5 Covariance and 95% confidence intervals

If $\mathcal{I}(\hat{\theta})$ is nonsingular and the local linear/Gaussian approximation is adequate, the estimator covariance is approximated by the inverse information matrix,

$$C_{\theta} = \text{Cov}(\hat{\theta}) \approx \mathcal{I}(\hat{\theta})^{-1}. \quad (19)$$

The marginal standard errors are

$$\text{SE}(\hat{a}_j) = \sqrt{(C_\theta)_{jj}}, \quad (20)$$

and the covariance-normalized parameter correlation is

$$\rho_{12} = \frac{(C_\theta)_{12}}{\sqrt{(C_\theta)_{11}(C_\theta)_{22}}}. \quad (21)$$

Using the standard-normal 97.5th percentile $z_{0.975} \approx 1.96$, a local marginal 95% Wald confidence interval is

$$\boxed{\hat{a}_j \pm 1.96\sqrt{(C_\theta)_{jj}}}. \quad (22)$$

These intervals are not clipped to the parameter box. If an estimate lies near an active bound, the sampling distribution is generally asymmetric and the unconstrained Wald approximation can be inaccurate.

5.1 Identifiability diagnostics

Let $0 \leq \lambda_{\min} \leq \lambda_{\max}$ be the eigenvalues of $\mathcal{I}$. The parameters are locally identifiable in the linearized problem precisely when $\mathcal{I}$ has rank two, equivalently when both eigenvalues are positive. Small $\lambda_{\min}$ indicates a weakly observed parameter combination. The spectral condition number

$$\kappa_2(\mathcal{I}) = \frac{\lambda_{\max}}{\lambda_{\min}} \quad (23)$$

measures anisotropy of the local information: a large value means that some parameter direction is much less informed than another. If the matrix is rank deficient, it cannot be inverted and finite standard errors or confidence intervals are unavailable without introducing additional information such as new observations or a prior/regularizer.

6 Numerical illustration

For the saved synthetic-data experiment with noise standard deviation $\sigma = 0.005$ and noise seed 42, the recovered coefficients were

$$\hat{\theta} = \begin{pmatrix} 0.7608972 \\ 0.6149885 \end{pmatrix}. \quad (24)$$

Central differences used the perturbations $h_1 = h_2 = 0.001$. The resulting Fisher information and covariance approximation were

$$\mathcal{I}(\hat{\theta}) = \begin{pmatrix} 4059.8200 & 2732.2310 \\ 2732.2310 & 3983.8231 \end{pmatrix}, \quad C_\theta = \begin{pmatrix} 4.57463 \times 10^{-4} & -3.13742 \times 10^{-4} \\ -3.13742 \times 10^{-4} & 4.66190 \times 10^{-4} \end{pmatrix}. \quad (25)$$

The information eigenvalues were approximately 1289.3263 and 6754.3168, giving numerical rank two and condition number 5.23864. The covariance implies correlation $\rho_{12} \approx -0.6794$: locally, an increase in one estimated diffusion coefficient can be partially compensated by a decrease in the other.

Parameter	Estimate	Standard error	95% lower	95% upper
a_1	0.760897	0.021388	0.718976	0.802818
a_2	0.614988	0.021591	0.572669	0.657308

The synthetic truth is $\theta_{\text{true}} = (0.8, 0.6)^\top$. Both true coefficients lie inside their corresponding approximate 95% intervals in this particular noise realization. This observation is a consistency check for one experiment, not evidence by itself that the procedure has exact 95% coverage over repeated samples.

7 Interpretive cautions

- The Fisher matrix and intervals are local quantities evaluated at $\hat{\theta}$. Strong nonlinearity can make the actual confidence region curved or asymmetric.
- The factor $1/\sigma^2$ assumes that the stated noise variance is known and that errors are independent and homoscedastic. With a general covariance Σ , the appropriate formulas are $\Phi = \frac{1}{2}q^\top \Sigma^{-1}q$ and $\mathcal{I} = S^\top \Sigma^{-1}S$.
- The calculation treats the forward map as exact. Mesh, time-discretization, and model-form errors can increase uncertainty or introduce bias beyond that captured by $\sigma^2 I_m$.
- Pointwise Wald intervals describe repeated-sampling uncertainty under the assumed statistical model. They are not Bayesian credible intervals and do not assign a posterior probability to a fixed parameter value.
- Simultaneous coverage of both parameters is not generally 95% when two separate marginal 95% intervals are reported. A joint confidence ellipse is instead based on $(\theta - \hat{\theta})^\top \mathcal{I}(\theta - \hat{\theta})$ and a chi-squared threshold with two degrees of freedom.

Summary

The weighted functional (7) is the negative Gaussian log-likelihood up to an additive constant. Central differences approximate the observation sensitivity, whose noise-scaled Gram matrix is the Fisher information (15). Inverting a full-rank Fisher matrix gives the local covariance approximation (19), from which the marginal 95% Wald intervals (22) follow.